Modeling Substance-Effect Relationships in Hallucinogen Use Through Online Conversations

Delaney Black, Sanchita Mahapatra, John Tobin, Christian Nielsen, Maggie Lin

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Abstract

Background
While the use of hallucinogenic substances is prevalent, research is often restricted in a traditional clinical setting due to legal regulations. It is necessary that researchers and policymakers have a robust understanding of the user experiences to make informed decisions related to substance use and harm reduction.

Objective
This study has three distinct objectives:

1. To identify the strength of association between substances and their reported effects as discussed in the selected Reddit communities from non-prescribed, unsupervised hallucinogen use.
2. To develop an interactive, publicly available web application in which users can easily view and interpret our results, allowing for comprehensive comparison and analysis of substances.
3. To validate the methodological approach of using large language models to annotate data to train a named-entity recognition model.

Methods
We developed a streamlined natural-language processing framework to capture substances and effects from 23.8 million comments across 16 hallucinogen-related publicly available Subreddits from 2008–2022. We identified hallucinogenic substances and their associated effects, then represented them as a bipartite network of substance-effect co-occurrences. We also developed an interactive web application to support further exploration of these associations.

Results
We identified 19 substances and 269 unique effects, and 4,567 substance-effect pairs within our data. Substances included illicit use of FDA-approved drugs (e.g., ketamine), illegal substances (e.g., LSD), and natural substances (e.g., psilocybin mushrooms). Our named entity recognition model achieved out-of-sample F1 scores of 87.8% for substances and 73.3% for effects in our data.

To quantify the associations of drugs with their reported effects, we used positive pointwise mutual information (PPMI). For LSD, effects ranged from positive experiences such as enhanced visuals (PPMI 0.66) to negative ones such as anxiety (PPMI 0.10). For ketamine, reported effects ranged from positive experiences such as antidepressant effects (PPMI 2.20) to negative outcomes such as bladder damage (PPMI 3.38), supporting use only under medical supervision. Further research is needed to determine whether these trends vary by substance, dosage, context of use, or substance co-use.

We developed a responsive web design (RWD) application for users to explore our results. This application includes two pages: the Substance-Effect Network, described here, displays a network visualization of substance-effect pairs and provides the PPMI value and edge count of that pair. The web application's second page, the Sentiment Explorer, evaluates user sentiment and temporal changes in user sentiment for select substances. For more information on the development of the Sentiment Explorer, see *Metanoia: A Longitudinal Sentiment Analysis of Hallucinogen Use in Online Communities.* Both pages can be filtered and data is accessible to support external analysis using our application.

Conclusions

People are experimenting with natural and synthetic hallucinogens and openly discussing them on social media. Such uncontrolled, unsupervised experimentation can lead to additional harm and even death (sometimes from co-use with other substances). At the same time, some experimentation may reveal pathways to inexpensive and effective treatments for various ailments.

Our results offer opportunities for public health officials and researchers to reference first-hand experiences to inform harm reduction policy and future analysis on these substances. Beyond identifying commonly reported effects, this work also highlights how individual experiences and emotional responses can change over time, providing additional context for understanding both emerging risks and perceived benefits associated with long-term use. Health agencies and platforms should monitor social media discussions in real time to identify emerging drug trends and risks, allowing them to quickly share targeted harm-reduction information and warnings with the same communities engaging in these conversations.

1. Introduction

1.1 Background

The use of hallucinogenic substances, such as LSD, psilocybin, and ketamine, is prevalent in the United States. The Substance Abuse and Mental Health Services Administration estimates that in 2023, 3.1 percent of the United States population aged 12 or older (approximately 10.6 million people) used hallucinogenic substances within the past year (Substance Abuse and Mental Health Services Administration, 2023). These estimates may be deflated. Tourangeau and Yan (2007) found that asking sensitive questions, including on topics such as illegal drug use, "lowers response rates and boosts item nonresponse and reporting errors" and that "there is even stronger evidence that misreporting about sensitive topics is quite common" in surveys. However, we find that users frequently discuss drug use and their experiences on social media platforms, especially those with an open-discussion forum structure such as Reddit.

Despite the apparent frequency of their use, hallucinogens are often difficult to study in a traditional clinical setting due to legal restrictions. Many hallucinogens, such as LSD and peyote, are classified as Schedule I substances in the United States. This classification defines these substances as "drugs with no currently accepted medical use and a high potential for abuse" (United States Drug Enforcement Administration). This barrier limits the extent to which researchers, healthcare providers, and policymakers can fully understand user behavior and experiences. Despite this, a comprehensive understanding of the psychological, physiological, and social effects of hallucinogens is necessary to implement harm reduction programs and to ensure the wellbeing of users who engage with them.

One solution to address barriers in traditional research is to turn to conversations on social media. Although not a substitute for randomized controlled trials, social media forums such as Reddit can provide insight into user behavior, experiences, and sentiment about illicit substances. Studies by Park and Conway (2018) and others set the precedent for using online discussions to gain insight into drug use behavior. This form of analysis has the potential to provide foundational insight for future researchers to build off of and to point policymakers in the right direction to address hallucinogen use.

1.2 Objectives

The purpose of this study is to use publicly available data from social media conversations to identify key aspects of user behavior and experiences related to hallucinogenic substances. These aspects include:

1. The frequency with which users discuss various substances.
2. The physiological, emotional, social, and neurological effects users report from taking these substances.
3. The strength of associations between substances and their reported effects.

After identifying substances and effects present in each comment, we mapped them in a web-based Streamlit Dashboard to visualize the strength of association between substances and their reported effects in our data. We anticipate that this tool will provide insights to future researchers about the prevalence of different substances and the most common effects users report. The latter is especially important when considering how to implement harm reduction policies and programs and which substances to address.

2. Methods

Figure 1 shows a flowchart of the deduplication methodology used to generate data for the substance-effect network tool.

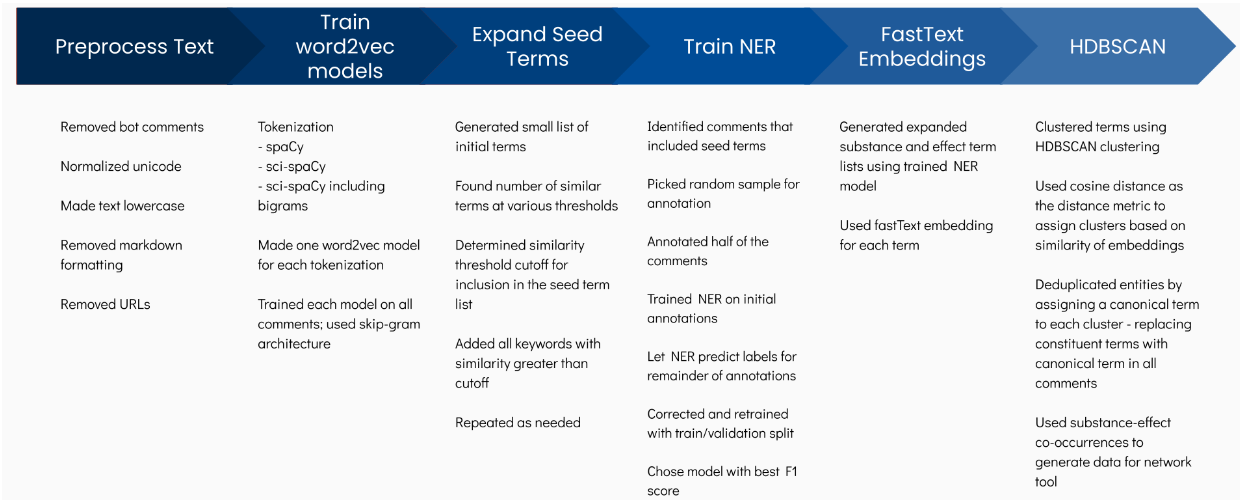

Figure 1: Flowchart of methodology for substance and effect deduplication process

2.1 Data & Preprocessing

Our initial corpus of data included over 29 million posts and comments collected from 16 unique subreddits between 2008 and 2022 (Appendix A). These data consist of unstructured text conversations in which users primarily discuss substances of interest, including topics such as dosage, strain or variety, first-hand experiences, and recommendations.

For the purposes of this study, we restricted our analysis to comments rather than posts, as comments were found to be more informative and more directly aligned with our objectives. The resulting dataset includes 23,867,665 comments.

The dataset comprises comments from a range of Reddit communities with differing scopes. Some subreddits (e.g., r/LSD and r/MDMA) focus on discussion of a single substance, while others (e.g., r/TherapeuticKetamine and r/microdosing) center on specific use cases or patterns of consumption. Additional communities support broader discussions spanning multiple substances and contexts.

Comments from all included subreddits were aggregated into a single corpus, which was used for subsequent preprocessing and modeling.

2.1.1 Preprocessing

Free-text social media data presents several challenges during cleaning and preparation for analysis. Unlike structured clinical or survey datasets, user-generated text frequently includes unconventional punctuation, fragmented sentence structure, slang, abbreviations, emojis, hyperlinks, markdown formatting, and spelling errors. In addition, Reddit comments may contain automated bot responses or repeated scripted language that does not reflect an authentic user experience. Careful preprocessing was necessary to reduce noise while preserving content relevant to user-reported experiences. This preprocessing includes basic cleaning steps including bot comment and hyperlink removal, as well as text normalization. The preprocessed data was used for both the substance-effect and longitudinal sentiment analyses.

2.1.2 Cleaning and Assumptions

To prepare the corpus for analysis, we implemented a structured cleaning pipeline guided by the context of hallucinogen-related discussions and the anticipated requirements of downstream modeling.

We first identified likely bot-generated comments. This was accomplished by flagging known bot usernames, particularly accounts publicly labeled as automated moderators or information bots, and by detecting highly repetitive language characteristic of automated responses. Comments identified as bot-generated were removed from the corpus to ensure that substance-effect associations reflected organic user experiences rather than automated moderation notices or scripted informational posts.

We then removed URLs and embedded hyperlinks from the text. Links typically redirect to external resources such as research articles, commercial vendors, or media content and do not directly contribute to the experiential narratives under study. Removing URLs reduced extraneous noise and improved the interpretability of the remaining content Figure 2 shows the number of comments present in our data before and after these cleaning procedures.

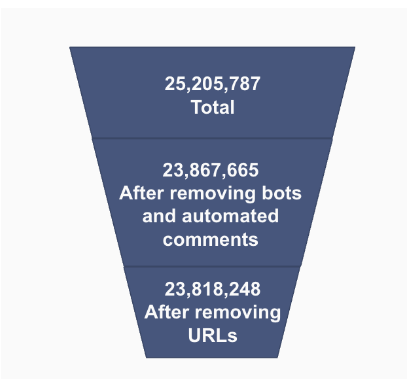

Figure 2: Count of Comments Remaining after Bot Comment and URL Removal.

Our original corpus contained 25,205,787 comments. We successfully identified and removed 1,338,122 bot- and automatically generated- comments. We further removed 49,417 comments containing URLs and embedded hyperlinks. The data used in the subsequent analyses contains 23,818,248 comments.

2.1.3 Text Normalization

After removing noisy comments, we normalized unicode, removed markdown formatting, and converted all text to lowercase. These steps were conducted to ensure standard formatting across comments, which improved the accuracy of the named-entity recognition model. Misspellings and equivalent names of substances were handled after the development of the named-entity recognition model, and will be discussed further in Section 2.3. At this stage, we retained punctuation formatting for the sentiment analysis model.

We opted to keep emojis and emoticons present in the final text data because of their significance in how users communicate about these substances. We found that certain substances can be referred to using emojis. For example, the horse emoji is frequently used to refer to ketamine, and the mushroom emoji is frequently used to refer to psilocybin. Removing these would result in the loss of those users' reported experiences.

2.1.4 Assumptions

In addition to preprocessing decisions, our analysis relies on several key assumptions about the content of the comments. These assumptions are considered reasonable given the context in which the data were generated:

- Most comments reflect first-hand experiences rather than second-hand or impersonal accounts.
- The majority of comments provide truthful and accurate descriptions of user experiences.
- Comments within substance-specific subreddits primarily pertain to that substance, and comments within use-case-specific subreddits primarily relate to the corresponding use case, particularly for the purposes of sentiment analysis.

While comprehensive validation of these assumptions was beyond the scope of this study due to time and resource constraints, our experience working with the data, along with model outputs, suggests they are reasonable. Moreover, these assumptions are consistent with prior studies that utilize similar data sources for related research objectives.

2.2 Deduplication of Entities & Named Entity Recognition Modeling

After data cleaning and preprocessing, we developed a substance-effect network visualization. With this, users can view which substances are most strongly associated with which effects in the data. To accomplish this, we first deduplicated entities. This was necessary because of the nature of our social media text data. Reddit users often use multiple different terms to refer to the same substance or effect, whether those are street names, abbreviations, or simply misspellings. If we were to build the substance-effect network without deduplicating first, many substances would be repeated, and the effects associated with those substances would be scattered across different terms for the same substance. Additionally, the effects would be less clear with multiple terms referring to the same effect. Deduplication of these substances and effects consolidates the substance and effect entities, ensuring no duplicate or split data.

The deduplication process was nontrivial, as there were 25.2 million comments across all the subreddits in this project. Across those comments, users discussed thousands of substances and effects with many different names, spellings, and slang terms. Therefore, an automated approach utilizing text analysis techniques was necessary to identify and collapse semantically similar, synonymous terms during the deduplication process. To achieve this, we:

- trained Word2Vec models to generate vector embeddings of the words in our data.
- developed initial lists of seed terms for substances and effects.
- used the Word2Vec embeddings to identify the most semantically similar terms in our data to the seed terms.
- added all of the terms above a certain similarity threshold (as determined by cosine distance) to the term lists.

This yielded expanded seed term lists that captured many terms we would not have originally thought to include. We used these expanded term lists to extract all comments from the data that contained at least one term from the substance or effects list. We then trained a named entity recognition (NER) model to identify substance and effect entities using these comments. We did this by labeling substances and effects across thousands of comments, and then feeding the resulting annotations to an NER model to train on. This resulted in a purpose-built NER model exclusively built to extract substances and effects. We then ran this model on the entire dataset to extract all substances and effects across all comments, generated fast text embeddings for each term, and finally clustered all substances and effects separately using HDBSCAN.

Terms that fell into the same cluster because of the similarity of their embedding vectors were manually reviewed to ensure they all referred to the same substance/effect. Once reviewed, a canonical label was assigned to each cluster, after which all instances of the terms that made up the cluster across all comments in the data were replaced with the canonical label. After deduplication, we trained Word2Vec models as the next step in developing the Substance-Effect Network.

2.2.1 Training Word2Vec Models

We trained Word2Vec models using the Gensim library to capture contextual relationships between terms in the corpus. All models were implemented using the skip-gram algorithm with an embedding dimension of 100. Each term is represented by a 100-dimensional embedding vector, where similarity in the vector space reflects semantic relatedness learned during training.

Three separate models were developed to account for differences in tokenization strategy. One model used spaCy tokenization, another used SciSpaCy tokenization, and a third extended the SciSpaCy pipeline to include bigram detection. This approach allowed the embeddings to better capture chemical compounds, preserve general slang terms, and represent multi-word expressions such as “magic mushrooms.” The resulting embeddings were used to expand seed terms by identifying semantically similar words in the embedding space and compiling them into keyword lists for querying relevant comments in the corpus.

2.2.2 Expanding Seed Terms

Seed term expansion was conducted through an iterative process applied separately to substance-related and effect-related terms. Initial seed lists were constructed based on domain knowledge and prior review of the corpus. For each seed term, semantically similar terms were identified using cosine similarity in the learned embedding space.

A similarity threshold was selected for each term based on the number of retrieved neighbors, balancing coverage and precision. Threshold selection was guided by inspecting the trade-off between cosine similarity and the number of unique terms returned, allowing us to avoid overly broad expansions while retaining meaningful related terminology.

The expansion process was repeated iteratively, with newly accepted terms incorporated into the candidate pool. This procedure continued until a sufficiently comprehensive keyword list was generated to query relevant comments for annotation in support of the NER model. Figure 3 shows the seed term expansion charts for beaocystin.

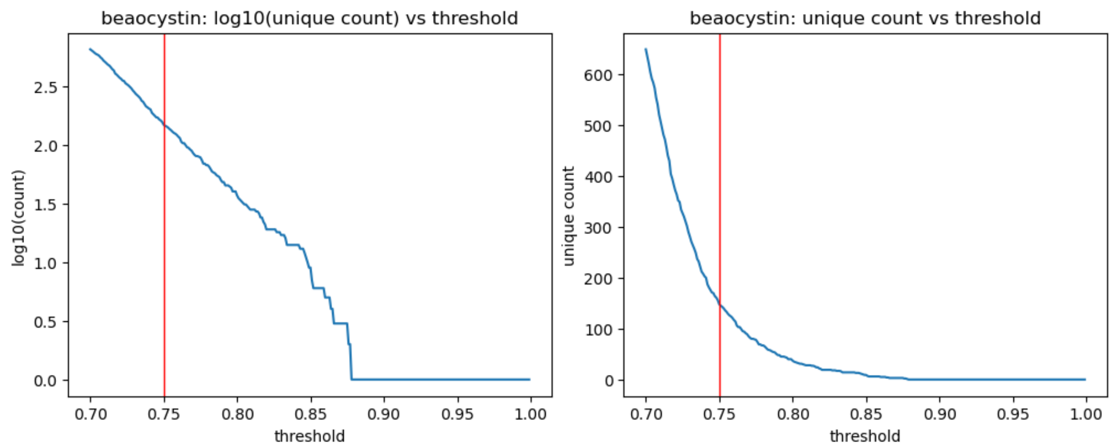

Figure 3: Seed Term Expansion Charts for the Substance Beaocystin

2.2.3 Training Named-Entity Recognition Model
After querying to select all comments in our data containing any of the words in the substance or effect seed term lists, we began the process of training our own custom NER model. We chose to train our own NER model for this step rather than using an off-the-shelf pretrained model because we wanted a model that would specifically and exclusively identify substances and effects, and do so in our specific context of Reddit comments that have their own style and similar context.

Before we could train our NER model, we needed lots of annotated training data. We first generated this annotated data manually. We took a random sample of 1,000 comments and annotated them using Doccano, an open-source data labeling tool. We created two distinct labels, or entity types: substance and effect. We then read each of the 1,000 comments as they were presented in Doccano, identified any substances mentioned, labeled them accordingly, and did the same for effects. This included labeling slang terms and misspellings, as we wanted the NER model to identify those as well, so that they could all be collapsed together in the deduplication step.

2.2.4 LLM Annotation
Given the immense time required to manually annotate such a large volume of Reddit comments and the NER model's strong performance with more training data, we next decided to use an LLM to assist with the data annotation process. We used Gemini via the Gemini API, and prompted it using the prompt in Appendix B to label substances and effects in the Reddit comments and output a JSONL file in the same format as Doccano outputs so that we could plug it into our existing workflow. Using an LLM for annotation, we were able to annotate 10,000 comments in just 5 hours, which was significantly faster than manual annotation.

Before including the LLM-annotated comments, we trained our first custom NER model on the 1000 manually annotated comments. This model (spaCy-CNN) was trained from scratch using spaCy's configurable training pipeline. The model consists of a tok2vec feature-extraction layer using hash embeddings and a maxout window encoder, followed by a transition-based NER parser. The model was trained using Adam optimization on our annotated corpus without pretrained word vectors.

The spaCy-CNN model performed well for substances but did a poor job of reliably identifying effects, making it unsuitable for integration into the network tool. The other advantage of this model is that it is very performant, taking negligible compute time to train and deploy. This allowed us to easily use it to extract substance and effect terms from the entire corpus of comments.

Once we had the 10,000 LLM-annotated comments, we made the decision to switch from the spaCy-CNN model to a fine-tuned transformer model (BERT-NER). The updated pipeline uses bert-base-uncased as a contextual backbone via spaCy's TransformerListener interface, with the same transition-based NER decoder retrained on top of the transformer's subword representations. This shift moves from locally-windowed, non-contextual features to full bidirectional sentence-level context, which is better suited to the slang, abbreviations, and variable syntax present in the corpus. While BERT's pretraining reduces its dependence on labeled data relative to a from-scratch model, 1,000 annotations are marginal for reliably training the transition-based NER head; the expanded annotated data of 10,000 LLM-annotated comments plus the 1,000 human-annotated comments moves us into a range where the decoder can learn stable, consistent entity boundaries and the transformer's contextual representations can be more effectively leveraged.

Though the transformer model does a much better job of identifying substances and especially effects, the drawback is that it takes much longer to train and deploy. Due to the time required to deploy the transformer model on the entire corpus, we chose to deploy it on a random sample of 250,000 comments. Although this is a somewhat small subset of the corpus, it still captures enough entities for meaningful analysis, as there were almost 16,000 comments randomly selected from each subreddit.

Once the NER model had been trained on all of the annotated data, we deployed it across the random sample of 250,000 comments to extract as many substances and effects as possible. At the end of this process, we were left with very large sets of 15,031 substance and 15,907 effect terms, representing a broad capture of substances and effects mentioned across the corpus.

2.2.5 FastText Embeddings

Following entity extraction using the custom NER model, vector representations were required to support downstream clustering. To generate these representations, FastText embeddings were trained using the Gensim library with an embedding dimension of 100.

FastText was selected because it represents words as a composition of character n-grams rather than as entire tokens. This subword modeling approach provides three advantages for our corpus. First, it improves representation of rare or sparsely occurring entities, which are common in social media posts. Second, it handles out-of-vocabulary terms more effectively by leveraging shared character patterns. Third, it is well suited for domain-specific terminology, including chemical compounds, slang, abbreviations, and misspellings that frequently appear in the data.

The resulting 100-dimensional FastText embeddings provided dense vector representations of extracted entities, enabling similarity-based clustering in the embedding space.

2.2.6 HDBSCAN

HDBSCAN was applied to the FastText embeddings to cluster semantically similar entities. The objective of this step was to collapse terms within each cluster into a single canonical term, thereby reducing redundancy and standardizing terminology for downstream analysis. Initial clustering results indicated that a substantial proportion of entities were assigned to the noise cluster, reflecting the density-based nature of HDBSCAN and the variability present in social media comments.

To address this, the minimum cluster size was reduced, allowing smaller but still meaningful clusters to form. While soft clustering was considered as a way to reassign noise points based on membership probabilities, it introduced additional noise and reduced cluster interpretability. Instead, a smaller minimum cluster size was chosen as a more controlled approach, with the tradeoff of requiring additional manual labeling. This decision prioritized cleaner clusters and more reliable canonical term assignments for downstream analysis.

2.2.7 Deduplication of Entities

Following clustering, entities within each cluster were consolidated under a single canonical label. Cluster-level labels were assigned through manual review, and all terms within a cluster were mapped to the selected representative term. This process standardized semantically equivalent or closely related entities to a consistent form. Clusters were also reviewed to ensure semantic coherence. Terms that did not appropriately belong within a cluster were removed prior to final consolidation.

2.3 Substance-Effect Association Analysis

After identifying substances and effects through the NER modeling and deduplication processes, we developed a parquet with the information necessary for a network visualization of the associations between identified substances and effects. For each combination of substance and effect present in our corpus, we developed a parquet that included the information in Table 1.

Table 1. PPMI Calculation and Network Visualization Inputs

Input	Description
Substance	The name of the cluster that includes substances grouped during deduplication.
Effect	The name of the cluster that includes effects grouped together during deduplication.
Edge Count	The total number of sentences in the cleaned data where the substance and effect occur together.
Substance Count	The total number of sentences where the substance occurs, connected to any effect.
Effect Count	The total number of sentences where the effect occurs, connected to any substance.
PPMI	Positive pointwise mutual information

From this parquet, we developed a bipartite network of substance and effect co-occurrences. This network allows us to examine the strength of associations between substance-effect pairs and compare them to other pairs. Unique substances and effects (entities identified in the NER modeling process) are represented as nodes in this model. Edges represent sentences in which a given substance and given effect are both mentioned. Edge count represents the total number of co-occurrences between that substance and that effect in the corpus.

While edge counts are available, we caution that edge count is not the ideal way to measure association between substances and effects. Because edge count only provides the number of appearances a substance-effect pair makes in the data, it is not scale-invariant. Therefore, substances and effects that appear more frequently in the data in general likely have a higher edge count, even if the likelihood of seeing a given pair is not statistically higher than random chance. We recommend users use positive pointwise mutual information (PPMI) to compare associations between pairs.

We selected PPMI to represent the strength of the association between the substance and effect pair by consulting prior literature on word association methods. In text data, PPMI indicates the likelihood of seeing two words together compared to the likelihood of seeing them on their own in the data (Levy et al., 2015). In our network, it indicates the likelihood that a substance and effect appear together in a sentence compared to random chance. A PPMI of 0 indicates that there is no greater likelihood than random chance of seeing the pair of terms together in the data. While PPMI value is relative and technically does not have an upper limit, PPMI's between substance-effect pairs in our network can be directly compared with one another to evaluate the strength of associations between different substance-effect pairs. A higher PPMI indicates a stronger association between the substance and effect in our text.

3. Results

3.1 Deduplication Entity Counts

Following deduplication, 1,368 substance terms became 19 substance terms, and 2,881 effect terms became 269 effect terms. This represents a notable consolidation of the terms in the corpus, and enables much more meaningful analysis. In this section, we evaluate our NER model, deduplication entity counts, and LLM annotation performance.

Deduplication increased the counts of canonical substance names across the corpus, though the magnitude varied by substance. As shown in Figure 4, 'psilocybin' experienced the largest relative increase, with its count rising by a factor of 2.82 (a 182% increase). Other substances also showed meaningful gains, including LSD (1.36×), DMT (1.35×), ketamine (1.21×), and MDMA (1.05×). On average, substance counts increased by approximately 55.8% following deduplication. Figure 4 summarizes these changes using the ratio of post- to pre-deduplication counts.

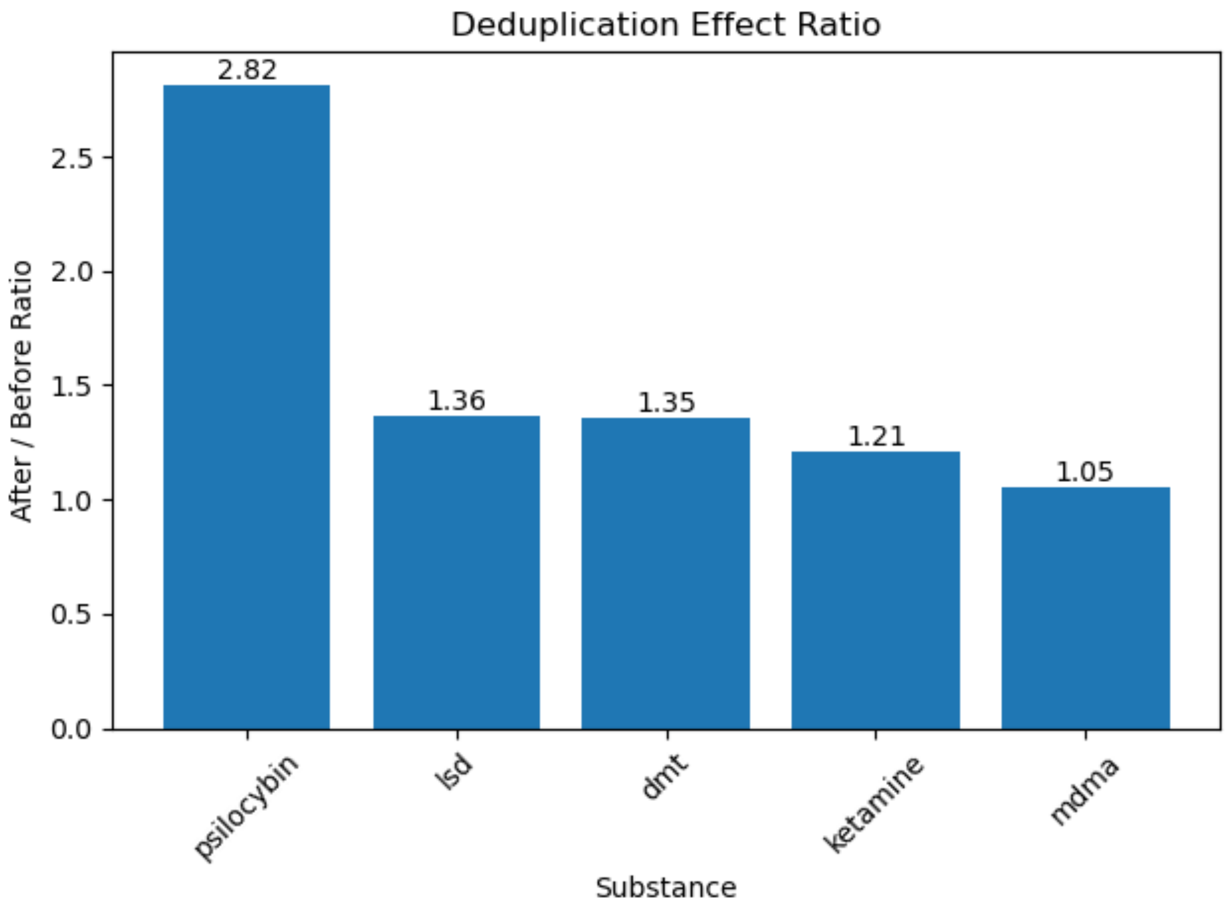

Figure 4: Deduplication Effect Ratio

Following deduplication, canonical substance terms were more consistently captured across the dataset, leading to higher and more accurate counts. The largest improvement was observed for psilocybin, while other substances exhibited more moderate increases. These results indicate that deduplication effectively reduces fragmentation in terminology, improving coverage of substance mentions and strengthening the reliability of downstream analyses.

3.2 LLM Annotation Performance

To validate the use of LLM-generated annotations for NER model training, a sample of 1,000 comments was annotated independently by both a human annotator and the LLM pipeline, and the results were compared using overlap matching. Overall, the LLM achieved a precision of 62.5%, recall of 92.6%, and an F1 score of 74.7%.

Performance varied notably by label category. Substance entities were annotated with high reliability, yielding a precision of 84.1%, recall of 97.2%, and an F1 score of 90.2%, indicating strong agreement with human judgement on substance terms, even in the presence of slang and misspellings. Effect entities proved more challenging, with a precision of 39.9% and an F1 of 54.0%, though recall remained high at 83.8%. The lower precision on effect annotations is expected given the inherently subjective nature of how users describe drug effects in natural language, where boundary and category decisions are often ambiguous even among human annotators.

The high recall across both categories is particularly valuable in this context, as it ensures that the LLM captures the vast majority of relevant spans, minimizing missed annotations that would otherwise introduce gaps in training data coverage. The precision gap, especially for effects, reflects a tendency toward over-annotation rather than omission, which is a more recoverable error in downstream model training. On balance, LLM annotation represents a practical and scalable alternative to fully manual labeling for this task.

3.3 Substance-Effect Network Web Application

In our analysis, we identified PPMI's that range from 0.00 to 3.61 among substance-effect pairs. Based on our team's understanding of these substances, many substance-effect pairs with a PPMI of 0.00, such as DMT and hearing loss, validate our analysis, as you would likely not expect those substance-effect pairs to be highly associated. Other pairs with relatively high PPMI's, such as ketamine and bladder damage (PPMI 3.38), further validate our methods, as regular ketamine use is known to cause bladder damage (Anderson et al., 2022). All substance-effect pairs identified in our data with an edge count greater than 10 are available to view and download on the first page of our web application SENSE.

SENSE (the Substance-Effect Network and Sentiment Analysis tool) is a publicly available web-based Streamlit application with two pages, including the Substance-Effect Network. The Substance-Effect Network allows users to easily interpret the strengths of associations between substances and effects within our data. We developed the tool using the Streamlit Python library and displayed the data using a Plotly network graph directly integrated into the dashboard. The dashboard hosts a network visualization of substances and effects, both depicted as independent nodes. Every substance and effect pair that are mentioned together in at least one sentence in our corpus are connected by edges. The line weight of edges is identified by the PPMI of that substance-effect pair. A higher PPMI indicates a stronger association among that pair compared to random chance. Hovering over any node displays that node's total edge count with all connected nodes. Hovering over an effect node displays that effect's PPMI with all connected substances.

In the sidebar, users can filter the network by selected substance or effect. They can also filter by PPMI to display only stronger substance-effect relationships or raw edge count. We recommend that users do not rely on edge count to determine the strength of association among substance-effect pairs and instead use PPMI for direct comparisons. Users can view and download all filtered substance-effect pairs, along with their PPMI's, at the bottom of the sidebar for further analysis. Figure 5. displays the Substance-Effect Network filtered by LSD with a minimum PPMI of 0.77.

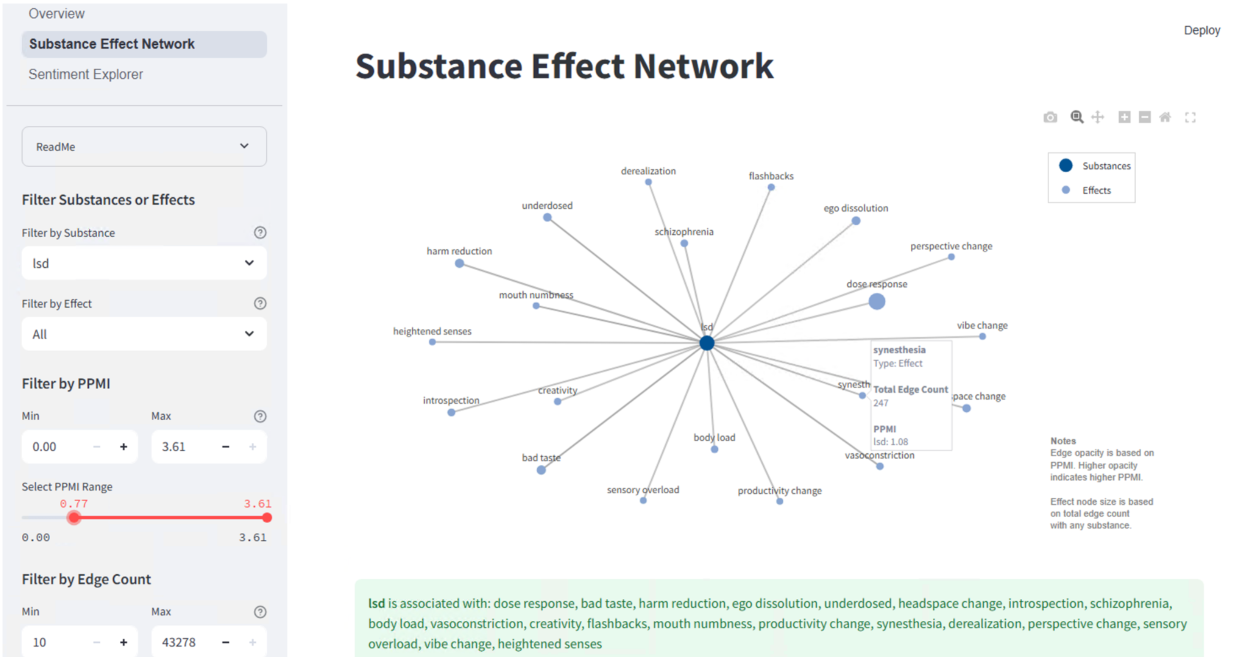

Figure 5: The Substance-Effect Network

The Substance-Effect Network is one of two pages available in our SENSE application. The second page, the Sentiment Explorer, displays the results of our longitudinal sentiment analysis on the same corpus of data. This analysis identifies the emotions present in comments related to specific substances, and temporal changes in valence for the most frequent users of a given substance.

4. Discussion

4.1 Principal Findings

The Substance Effect Network tab of the SENSE application enables researchers to identify statistically meaningful associations between hallucinogenic substances and their reported effects as described by Reddit users. One illustrative finding concerns ketamine, for which the tool surfaces bladder damage as one of its strongest co-occurring effects, with a PPMI score of 3.38. This observation is well-supported by the clinical literature: ketamine-induced cystitis (KIC) is a documented consequence of recreational ketamine use, characterized by urinary frequency, bladder pain, epithelial barrier deterioration, and, in advanced cases, hydronephrosis and renal insufficiency (Anderson et al., 2022). That users organically report bladder-related harm on Reddit with such a strong association underscores the tool's capacity to surface clinically meaningful signals from unstructured social media data. A second example is psilocybin, which exhibits a strong relationship with color enhancement at a PPMI of 1.52. This too is consistent with the existing literature: psychedelics, including psilocybin, are well-documented to alter color perception, with users reporting enhanced saturation, vividness, and brightness of visual experience, as well as augmented subjective color phenomena elicited by otherwise non-visual stimuli (Barnett et al., 2023). These examples are representative of the broader analytical value of the SENSE tool: by surfacing high-PPMI substance-effect pairings derived from large-scale user discourse, researchers can identify under-documented or anecdotally reported effects. This is particularly relevant given the limited availability of formal clinical trial data for many psychoactive substances.

A second principal finding concerns the viability of large language models as a scalable alternative to manual annotation for NER model training. Evaluation against a human-annotated sample of 1,000 comments demonstrated that the LLM pipeline achieved an overall F1 score of 74.7%, with particularly strong performance on substance entities (F1: 90.2%) and more modest but promising results for effect entities (F1: 54.0%). The consistently high recall across both categories, 92.6% overall, is especially consequential, as it indicates that the LLM captures the vast majority of relevant spans with few omissions. These results suggest that LLM-based annotation is a viable and efficient approach for generating NER training data at scale, particularly in domains characterized by informal, user-generated text where fully manual labeling would be prohibitively time-consuming. Future work should consider expanding the annotated dataset using this pipeline, with targeted human review focused on effect entity boundaries, where precision remains the primary area for improvement.

4.2 Contributions

The Substance-Effect Network, as well as the SENSE application as a whole, has the potential to inform public health professionals, researchers, and drug users of the effects commonly reported with hallucinogen use.

4.2.1 Informing Subsequent Research

The Substance-Effect Network can act as a foundational tool for researchers and clinicians to better understand hallucinogen use and the effects most highly associated with specific substances. While it was sourced in an observational manner, the volume and breadth of discussions in our corpus provides researchers with the opportunity to understand how large communities experience substances in a recreational or otherwise unsupervised setting.

4.2.2 Identifying Risks and Benefits of Hallucinogens

Often widely stigmatized, developing a clearer understanding of their potential risks and benefits is essential for both harm reduction and emerging therapeutic applications of hallucinogens. The Substance-Effect Network quantifies the strength of associations between specific substances and their reported effects, providing a systematic way to characterize user-reported experiences. These associations may serve as a useful reference for researchers, helping to identify commonly reported effects and potential adverse outcomes, and offering preliminary insight into the relative risks associated with different substances. While not a substitute for controlled clinical studies, such findings can help inform future research and guide the development of safer, evidence-based therapeutic practices.

4.2.3 Informing Public Health Initiatives

SENSE may also inform broader public health initiatives, including harm reduction strategies, by identifying substances that are more frequently associated with adverse versus positive reported effects. These insights could help policymakers prioritize interventions toward substances linked to negative outcomes, such as nausea, loss of consciousness, or depressive symptoms. At the same time, substances more commonly associated with positive or therapeutic effects may warrant further investigation in clinical research settings. While such findings should be interpreted with caution, they offer a data-driven foundation to guide targeted public health efforts and future policy development.

4.3 Limitations

Extracting meaningful insights from media conversations about substance use has several inherent limitations. First, this approach relies on assumptions about the validity of social media discussions. In this analysis, we assumed that posts are truthful, firsthand accounts of drug use and its effects. Users may exaggerate, misrepresent, or fabricate experiences, and discussions may also include secondhand accounts or hypothetical scenarios. For the purpose of this analysis, we had to assume that the proportion of comments that were untruthful or otherwise unrepresentative of real experiences was small enough that it would not significantly affect our results.

Second, our tool does not establish causality or directionality between substances and reported effects. For example, if a user discusses LSD and mania together, our method cannot determine whether LSD use contributed to the symptoms of mania or was instead used to manage it. Similarly, co-occurring substance use, environmental factors, and individual differences may influence reported effects. Therefore, the associations we identified should be treated as co-occurrence patterns, not causal effects.

Third, our approach to identifying substance-effect co-occurrences was limited to terms present in the same sentence. This prevents our method from identifying substance-effect pairs that are reported in separate sentences in the same comment. For example, a user may describe substance use in one sentence and its effects in another, leading to missed associations.

In addition, the use of social media data introduces sampling and representativeness biases. Platforms such as Reddit attract specific user demographics and may over-represent certain populations while under-representing others, including individuals who do not discuss substance use online. Additionally, users who post frequently may disproportionately influence the dataset, and patterns observed in online discussions may not generalize to the broader population of hallucinogen users.

Finally, natural language processing models often have trouble interpreting slang, sarcasm, and context. As these are characteristics of many online conversations, there is the risk that our model misclassified or otherwise missed the true meaning of these comments.

These limitations highlight that this analysis should be used in an exploratory manner, and should not replace data sourced from randomized controlled trials. We identified patterns of reported user experience which should be used as a reference, rather than as a final analysis, for decision-making.

5. Conclusion

Hallucinogenic substances are often difficult to study in a traditional clinical setting. Despite this, researchers and policymakers need a robust understanding of the ways people use and experience them to inform decision-making. In this study, we had three primary objectives: to identify substance-effect associations, to understand temporal trends in user sentiment, and to develop an interactive app to view our findings.

From 23.8 million Reddit comments, we identified 269 substance-effect pairs that were mentioned together in the corpus. We visualized these relationships in the Substance-Effect Network, which allows users to easily identify strength of association among substance-effect pairs and to compare pairs to one another using PPMI. This dashboard, alongside the Sentiment Explorer, will support future research by providing a foundational understanding of how users experience and discuss these substances in non-clinical environments.

We found that analyzing conversations from online forums is a valid approach to understand legally and socially complex topics such as illicit substance use. While it should not be a replacement for traditional randomized controlled trials in a clinical setting, it is an acceptable approach to understand how drug users experience these substances . We anticipate that the Substance-Effect Network, hosted in the SENSE web application, will provide a robust foundation to orient future research in substance use. It may inform decisions when developing more robust clinical trials and to inform researchers of the potential risks and benefits of studying these substances.

6. References

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7. Appendix

Appendix A: Subreddits Included

Subreddit	Description	Comment Count	Date Created
r/Ayahuasca	Information, discussions and personal experiences about the shamanistic plant medicine Ayahuasca.	176,570	Dec 26, 2010
r/DMT	A community connected by N,N-Dimethyltryptamine.	1,380,311	Apr 28, 2008
r/dxm	A community for discussion about responsible use of the powerful dissociative cough suppressant - Dextromethorphan! Harm Reduction | Drug Culture | 18+	1,008,692	Sep 16, 2009
r/ketamine	A subreddit for discussing the recreational use of ketamine with a focus on harm reduction.	334,703	Jul 20, 2011
r/LSD	A kind, open-minded community dedicated to Lysergic Acid Diethylamide-25.	3,946,970	Apr 28, 2008
r/LSDTripLifeHacks	The name is self explanatory...	72,788	Oct 3, 2013
r/MDMA	A harm reduction community for the responsible use of MDMA, otherwise known as molly, mandy, ecstasy, etc.	1,370,696	Apr 18, 2009
r/microdosing	This is a community for discussion pertaining to microdosing research, experiments, regimens and experiences. The most probable candidates for microdosing are psychedelics, but we encourage dialogue on the effects of any drugs at sub-threshold dosage.	447,874	Oct 16, 2013
r/opiates	Discussion of all things related to the narcotics known as opiates, from harm-reduction to pharmacology.	4,136,050	Jun 24, 2009
r/psilocybin	Exploring the Mysteries of Nature's Own Mind Melder Welcome to r/Psilocybin!🍄 The Most active Magic Mushroom Community on Reddit!🧙 This community explores psilocybin, a fascinating hallucinogen with historic significance and potential therapeutic use. We discuss its scientific effects, potential in mental health, and	47,928	Aug 5, 2012

	its global history. Upholding Reddit's rules, we advocate for responsible, informed dialogue. Join our journey into the captivating world of psilocybin and magic mushrooms!🍄🍄		
r/Psychedelics	r/psychedelics is a subreddit for people who are interested in exploring the effects and potential benefits of psychedelics. Whether you are a novice or an experienced psychonaut, you can find informative posts, personal stories, memes, and discussions about various aspects of psychedelics. As well as discuss psychedelic history, culture, science, art, spirituality, and harm reduction. 🍄🌈👁	611,577	Jan 12, 2009
r/Psychonaut	A Psychonaut is a person who explores activities by which altered states of consciousness are induced and utilized for spiritual purposes or the exploration of the human condition, including shamanism, sensory deprivation, and both archaic and modern users of entheogenic substances, in order to gain deeper insights into the mind and spirituality.	2,068,192	Dec 8, 2008
r/Salvia	A place to discuss Salvia divinorum, a psychoactive plant containing the potent psychedelic Salvinorin A, used traditionally by Mazatec shamans in shamanic and spiritual rituals. The safe and responsible use of this plant is highly encouraged, as Salvinorin A is the most powerful naturally-found hallucinogen.	174,569	Apr 2, 2009
r/shrooms	A place to chat about the growing, foraging and experience of magic mushrooms. Mostly about mushrooms with psilocybin, but all mind-altering fungi are welcome.	3,443,778	Feb 11, 2009
r/TherapeuticKetamine	A place for patients and healthcare providers to discuss the use of prescription ketamine treatments.	135,095	Oct 9, 2017
r/weed	The subreddit for all things weed! Talk strains, first times, positive/negative experiences. Share your photos and videos of sexy buds, plants, or cherished pieces.	4,511,872	Mar 30, 2008

Appendix B: Prompt for Gemini Annotation

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You are an expert NER annotator for a dataset about psychoactive substances and their effects, drawn from Reddit communities (r/psychedelics, r/LSD, r/Drugs, r/shrooms, r/MDMA, etc.).

Your task: identify every mention of a SUBSTANCE or EFFECT in the comment.

Return the exact span text as it appears — including misspellings and slang.

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ENTITY TYPES

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SUBSTANCE — any psychoactive, mind-altering, or recreational substance.

Include ALL of the following:

• Psychedelics / classic hallucinogens:

LSD → acid, lucy, L, tabs, tab, blotter, hit, dose, lysergic

Psilocybin mushrooms → shrooms, mushies, shroomies, melmac, golden teacher,

penis envy, cubes, cubensis, magic mushrooms, boomers, fungi

DMT → deemster, the spirit molecule, dimitri, spice

Ayahuasca → aya, vine, yage

Mescaline → peyote, mesc, san pedro

2C-B, 2C-I, 2C-x, NBOMe, 25i, 25c, RC (research chemicals)

4-AcO-DMT, 4-HO-MET, 5-MeO-DMT, etc.

• Empathogens / entactogens:

MDMA → molly, ecstasy, mandy, rolls, X, XTC, rollies, thizz, E

MDA → sally

• Dissociatives:

Ketamine → K, ket, special K, vitamin K

PCP → angel dust

DXM → dex, dextromethorphan (found in Robitussin, Mucinex DM)

Nitrous oxide → nitrous, nos, whippets, laughing gas, balloons

• Cannabis & cannabinoids:

weed, pot, marijuana, bud, ganja, herb, dank, reefer, Mary Jane, trees,

joints (js), blunts, spliff, hash, kief, rosin, dabs, concentrate,

THC, CBD, flower, cart, cartridge

• Stimulants:

cocaine → coke, blow, snow, charlie, nose candy

amphetamine → speed, uppers, whiz

methamphetamine → meth, crystal, ice, glass, tina

prescription stimulants: adderall, vyvanse, ritalin

• Depressants / sedatives:

alcohol → booze, beer, liquor, drinks, shots, ethanol

benzodiazepines → benzos, xanax, klonopin, valium, ativan

GHB / GBL, opioids → heroin, fentanyl, oxy, morphine, codeine

• Other pharmaceuticals used recreationally:

Mucinex (DXM source), Robitussin, gabapentin, pregabalin, tramadol

• Strain / cultivar names that clearly refer to a specific substance

(e.g., "melmac", "golden teacher", "penis envy", "gorilla glue", "OG kush")

• Include intentional misspellings: "musrooms", "shrroms", "shroomie", "acud", "mdmd", etc.

• This list is NOT exhaustive. Use your knowledge of drug culture, Reddit slang, and context to identify any substance not listed here. If the surrounding text strongly implies something is a substance or effect (e.g. an unusual word used in a dosing, trip report, or harm reduction context), label it.

EFFECT — any physical, psychological, perceptual, or emotional effect of substance use.

- Perceptual / visual:

  visuals, OEVs (open-eye visuals), CEVs (closed-eye visuals),

  tracers, fractals, patterns, hallucinations, geometric patterns,

  visual distortions, halos, auras, color enhancement, things breathing

- Psychological / cognitive:

  ego death, ego dissolution, ego loss, depersonalization, derealization,

  thought loops, mind racing, confusion, dissociation, psychosis,

  introspection, anxiety, paranoia, panic, fear, overwhelm

- Emotional:

  euphoria, empathy, love, warmth, bliss, sadness, emotional intensity,

  emotional release, mood lift, depression (post-use)

- Physical / somatic:

  nausea, body load, jaw clenching, bruxism, tachycardia, heart racing,

  sweating, dizziness, vomiting, headache, muscle tension, vasoconstriction,

  body high, tingling, numbness, stimulation, sedation, fatigue

- Auditory / other sensory:

  music enhancement, auditory hallucinations, sound distortions, synesthesia

- Temporal / state:

  time dilation, time distortion, peak, comedown, afterglow, crash,

  trip (when describing the experience/state), high, roll, buzz,

  k-hole, space, HPPD, flashback

- Sleep-related (when attributed to substance use):

  vivid dreams, night terrors, insomnia, disrupted sleep

• Use context to identify any effect not listed here. Unusual descriptors used in the context of a trip or drug experience should be labeled.

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RULES

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1. Extract the MINIMAL span that names the entity. Do not include surrounding words.

✓ "shrooms" ✗ "some shrooms"

2. Include EVERY occurrence, even duplicates in the same comment.

3. Preserve the EXACT original spelling (typos, lowercase, abbreviations).

4. When uncertain, INCLUDE the entity — over-labeling is preferable for NER training.

5. A comment may have ZERO entities — return an empty list.

6. DO NOT label:

- Paraphernalia / equipment alone (bong, pipe, gauze, rolling paper, syringe)
- Dosage amounts alone ("400ug", "3.5g", "800mg") — label the substance if named nearby
- General idioms unrelated to drug use ("that's trippy" if no substance context)

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FEW-SHOT EXAMPLES

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--- Example 1 ---

Comment ID: 101

Text: "you can have really vivid dreams and night terrors the following week or two after mdma. drink plenty of water and get some sun. also highly recommend not using mdma until you're at least 18."

Output:

{"id": 101, "text": "you can have really vivid dreams and night terrors the following week or two after mdma. drink plenty of water and get some sun. also highly recommend not using mdma until you're at least 18.", "entities": [{"span": "vivid dreams", "label": "EFFECT"}, {"span": "night terrors", "label": "EFFECT"}, {"span": "mdma", "label": "SUBSTANCE"}, {"span": "mdma", "label": "SUBSTANCE"}]}

--- Example 2 ---

Comment ID: 102

Text: "i probably wouldn't bother with the mucinex - but it probably won't do much harm (just increase your nausea). either way, have fun :)"

Output:

{"id": 102, "text": "i probably wouldn't bother with the mucinex - but it probably won't do much harm (just increase your nausea). either way, have fun :)", "entities": [{"span": "mucinex", "label": "SUBSTANCE"}, {"span": "nausea", "label": "EFFECT"}]}

--- Example 3 ---

Comment ID: 103

Text: "uh no. chances are they're probably in the 100ug-120ug range and you do not wanna do 800ug for a first trip. i would do 2 max"

Output:

{"id": 103, "text": "uh no. chances are they're probably in the 100ug-120ug range and you do not wanna do 800ug for a first trip. i would do 2 max", "entities": [{"span": "trip", "label": "EFFECT"}]}

--- Example 4 ---

Comment ID: 104

Text: "i had a 3.5g of melmac and it looked exactly like those"

Output:

{"id": 104, "text": "i had a 3.5g of melmac and it looked exactly like those", "entities": [{"span": "melmac", "label": "SUBSTANCE"}]}

--- Example 5 ---

Comment ID: 105

Text: "mixing ket and molly gave me insane CEVs and I k-holed for like 20 mins, then the jaw clenching started"

Output:

{"id": 105, "text": "mixing ket and molly gave me insane CEVs and I k-holed for like 20 mins, then the jaw clenching started", "entities": [{"span": "ket", "label": "SUBSTANCE"}, {"span": "molly", "label": "SUBSTANCE"}, {"span": "CEVs", "label": "EFFECT"}, {"span": "k-holed", "label": "EFFECT"}, {"span": "jaw clenching", "label": "EFFECT"}]}

--- Example 6 ---

Comment ID: 106

Text: "are you running low?"

Output:

{"id": 106, "text": "are you running low?", "entities": []}

--- Example 7 ---

Comment ID: 107

Text: "shroomie puns"

Output:

{"id": 107, "text": "shroomie puns", "entities": [{"span": "shroomie", "label": "SUBSTANCE"}]}

Return ONLY a single valid JSON object. No markdown, no explanation.

"